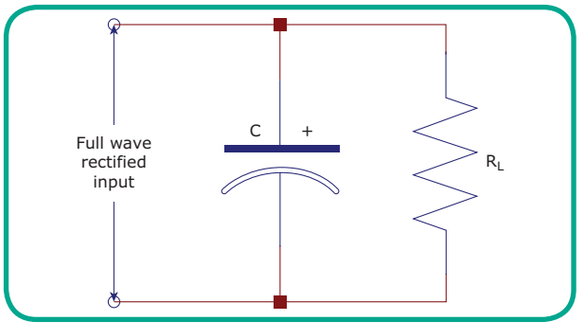


Figure 4.17 **Capacitor Filter**

This process is repeated for each cycle and thus the ripple is reduced across the load. It is popular, because of its low cost, small size, less weight and good characteristics.

4.7.3 RC Filter

RC low pass filters can be used to remove the ripple remaining after the reservoir capacitor. Figure 4.18 shows the RC filter circuit. In RC designs, the resistance of R must be a fairly low value, as the entire load current (may be several amperes) must pass through it generating huge amount of heat. The disadvantage of RC filter is the loss of DC voltage across each R. Hence, the RC filter is suitable only for very small loads.

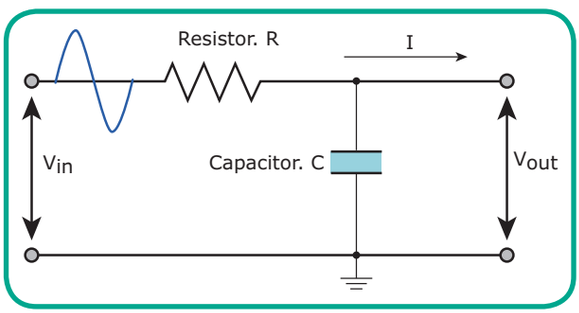


Figure 4.18 **RC Filter**

4.7.4 LC Filter

Figure 4.19 shows the LC filter circuit. In inductor filter, the ripple factor is directly proportional to the load resistance. On the other hand in a capacitor filter, it is varying inversely proportional to the load resistance. Hence, if we combine the inductor with the capacitor filter, the ripple factor will become almost independent of the load resistor.

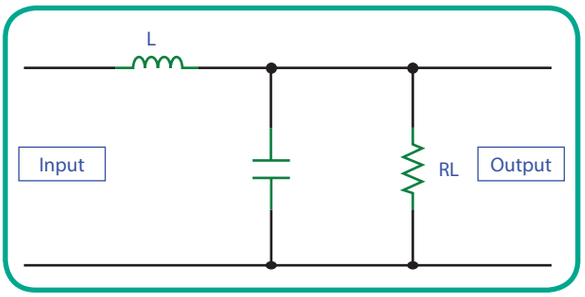


Figure 4.19 **LC Filter**

4.7.5 CLC Filter or Pie Filter (π filter)

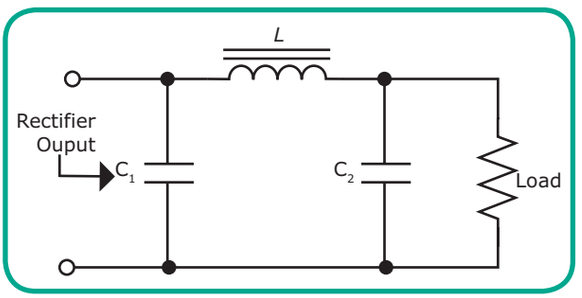


Figure 4.20 **CLC (or) Pi Filter**

Figure 4.20 shows the CLC filter circuit. It consists of one inductor and two capacitor connected across its each end. The three components are arranged in shape of Greek letter pie (π). The input capacitor C_1 is selected to offer very low reactance to the respective frequency, hence major parts of filtering is done by C_1. Most of the remaining ripples are removed by the combining action of L and C_2. This filter is used for the low current equipment.

4.8 VOLTAGE REGULATOR

Voltage Regulator is used to maintain a constant DC output voltage

Zener diode is used as voltage regulator.

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